KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 2nd Year 1st Term Examination, 2016
Department of Computer Science and Engineering
MATH 2107
Fourier Analysis and Linear Algebra

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define the following matrices with examples: (10)
(i) Upper triangular matrix, (ii) Row matrix, (iii) Null matrix, (iv) Unit matrix and (v) Real matrix

b) What is meant by a field? (10)

c) Verify that B is the inverse of A by showing AB = BA = I, where $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$. (05)

d) If $A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 1 & 0 \\ 2 & 3 & 1 & 2 \end{bmatrix}$ then find AB by partitioning. (10)

2. a) What is meant by symmetric and skew-symmetric matrices? Find the symmetric and skew-symmetric parts of the matrix $E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}$. (13)

b) What is meant by equivalent matrix? Find the rank of $\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$ by the elementary row transformation. (10)

c) Reduce $A = \begin{bmatrix} 6 & 3 & -4 \\ -4 & 1 & -6 \\ 1 & 2 & -5 \end{bmatrix}$ to echelon form and then to its row canonical form. Also find the rank of A. (12)

3. a) Apply rank test to solve the following system of linear equations: (12)

$$2x_1 + 3x_2 + x_3 = 9$$
$$x_1 + 2x_2 + 3x_3 = 6$$
$$3x_1 + x_2 + 2x_3 = 8$$

If consistent, then find its solutions.

b) Let $M = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$. Find all eigen values of M and the corresponding eigen vectors. Also determine an invertible matrix P such that $P^{-1}MP$ is diagonal. (13)

c) Test whether the following matrix P is invertible or not. If invertible, then find P^{-1}, where $P = \begin{bmatrix} 4 & 1 & -1 \\ 0 & 3 & 2 \\ 3 & 0 & 7 \end{bmatrix}$. (10)

4. a) Determine whether or not the vector v = (3, 9, –4, –2) is a linear combination of vectors e_1 = (1, –2, 0, 3), e_2 = (2, 3, 0, –1) and e_3 = (2, –1, 2, 1). (08)

b) Let W be the subspace of $\mathfrak{R}^4$ generated by the vectors (1, –2, 5, –3), (2, 3, 1, –4) and (3, 8, –3, –5). Find a basis and dimension of W. (10)

c) What is meant by orthogonal vectors? Examine whether the vectors (1, –2, 3, –4) and (5, –4, 5, 7) are orthogonal or not. (06)

d) Define inner product of a vector space and orthogonal of a vector space. If $X_1 = [1, 2, 3]'$ and $X_2 = [2, -3, 4]'$ find their inner product and length of each. (11)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Write down the assumptions for the validity of Fourier series expansion. Obtain Fourier series for the expansion of $f(x) = x\sin x$ in the interval $-\pi < x < \pi$, hence deduce that $\frac{\pi}{4} = \frac{1}{2} + \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \cdots$ (22)

b) If $f(t) = t^2,\ 0 \le t \le 1$. Find half range cosine series. (13)

6. a) Write down Parseval's identity of the Fourier series. Use parseval's identity to the function $f(x) = \sin x,\ 0 < x < \pi$ and show that $\frac{1}{1^2.3^2} + \frac{1}{3^2.5^2} + \frac{1}{5^2.7^2} + \cdots = \frac{\pi^2 - 8}{16}$. (20)

b) Find the Fourier cosine transform of $f(x) = \frac{1}{1+x^2}$ and hence find Fourier sine transform of $\varphi(x) = \frac{x}{1+x^2}$. (15)

7. a) A sinusoidal voltage $E\sin\omega t$, where t is time, is passed through a half-wave rectifier that clips the negative portion of the wave. Find the Fourier series of the resulting periodic function $u(t) = \begin{cases} 0 & ,\ \text{if } -\frac{\pi}{\omega} < t \le 0 \\ E\sin\omega t, & \text{if } 0 < t < \frac{\pi}{\omega} \end{cases}$. (18)

b) Find the Fourier integral of the function $f(x) = 0,\ \frac{1}{2}$ or e^{-x} for $x < 0, x = 0$ or $x > 0$ respectively. (17)

8. a) Define Z–transform. Determine the Z–transform of the signal $x(n) = a^n u(n) - b^n u(-n-1)$ $(a \text{ and } b) < 1,\ b > a$ and plot the ROC. (10)

b) Find the Z–transform of the signal given and discuss its properties: (15)

$$f(k) = \begin{cases} C^k, & k = 0, 1, 2, \cdots \\ 0, & k = -1, -2, \cdots \end{cases}$$

Where the constant C takes the following values (i) 0 < C < 1, and (ii) C > 1.

c) Find the inverse Z-transform of $X(Z) = \frac{Z}{(Z-1)(Z^2+1)}$ by using Residue method. (10)